📅 Day 18 Report – Introduction to Sniffing & Packet Analysis

Date: 05 July 2025
Module: Sniffing – Basics & Network Packet Capture
Mode: Theoretical + Practical

🔧 Main Focus

Introduced **sniffing attacks** and their role in interception of network traffic. Covered the theoretical fundamentals of **packet capture** and **network monitoring**.

📚 Theory Topics

- **What is Sniffing?**
  Capturing and monitoring data packets flowing across a network.
- **Types of Sniffing:**
  - **Passive Sniffing:** Monitors unencrypted traffic on hub-based networks.
  - **Active Sniffing:** Involves injecting traffic (e.g., ARP spoofing).
- **Protocols vulnerable to sniffing:**
  - HTTP, FTP, Telnet, POP3 (unencrypted protocols)

? Practical Tasks

- Installed **Wireshark**
- Captured live packets on local interface (eth0)
- Applied filters:
  - `http`
  - `ip.addr == 192.168.1.1`
  - `tcp.port == 80`
- Interpreted TCP handshake and DNS queries

🧠 Key Learnings

- Sniffing enables traffic interception on insecure networks
- Wireshark provides powerful insights into protocol behavior
- Network segmentation and encryption help prevent sniffing